Tema Alg Liniară

[redacted]

$$\begin{cases} 2x+y-z+t=1 \\ x-y+2z+t=3 \\ x+2y-3z-2t=6 \end{cases}$$

$$\left(\begin{array}{cccc|c} 2 & 1 & -1 & 1 & 1 \\ 1 & -1 & 2 & 1 & 3 \\ 1 & 2 & -3 & -2 & 6 \end{array}\right) \overset{L_2=L_3-L_2}{=} \left(\begin{array}{cccc|c} 2 & 1 & -1 & 1 & 1 \\ 0 & 3 & -5 & -3 & 3 \\ 1 & 2 & -3 & -2 & 3 \end{array}\right) \overset{L_1=L_1\cdot L_3}{=}$$

$$\left(\begin{array}{cccc|c} 1 & -1 & 2 & 3 & -2 \\ 0 & 3 & -5 & -3 & 3 \\ 1 & 2 & -3 & -2 & 3 \end{array}\right) \overset{L_3=L_3-L_1}{=} \left(\begin{array}{cccc|c} \boxed{1} & -1 & 2 & 3 & -2 \\ 0 & 3 & -5 & -3 & 3 \\ 0 & 3 & -5 & -5 & 5 \end{array}\right) \begin{array}{l} L_3=L_3-L_2 \\ L_2=\frac{1}{3}L_2 \\ L_3=-\frac{1}{2}L_3 \end{array}$$

$$\left(\begin{array}{cccc|c} 1 & -1 & 2 & 3 & -2 \\ 0 & 1 & -\frac{5}{3} & -1 & 1 \\ 0 & 0 & 0 & -2 & 2 \end{array}\right) \overset{L_1=L_1+L_2}{=} \left(\begin{array}{cccc|c} 1 & 0 & \frac{1}{3} & 2 & -1 \\ 0 & 1 & -\frac{5}{3} & -1 & 1 \\ 0 & 0 & 0 & -2 & 2 \end{array}\right) =$$

$$\left(\begin{array}{cccc|c} 1 & 0 & 1/3 & 2 & -1 \\ 0 & 1 & -5/3 & -1 & 1 \\ 0 & 0 & 0 & 1 & -1 \end{array}\right) \overset{L_2 = L_3 + L_2}{=\!=} \left(\begin{array}{cccc|c} 1 & 0 & 1/3 & 2 & -1 \\ 0 & 1 & -5/3 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 \end{array}\right) \overset{L_1 = L_1 - 2L_3}{=\!=}$$

$$\left(\begin{array}{cccc|c} \boxed{1} & 0 & 1/3 & 0 & 1 \\ 0 & \boxed{1} & -5/3 & 0 & 0 \\ 0 & 0 & 0 & \boxed{1} & -1 \end{array}\right) \Rightarrow$$

sistem compatibil nedeterminat

x, y, t necunoscute principale

$z = \alpha$ necunoscută secundară

$x + 1/3\,\alpha = 1 \Rightarrow x = 1 - 1/3\,\alpha$

$y - 5/3\,\alpha = 0 \Rightarrow y = 5/3\,\alpha$

$t = -1$

$$Sol = \left\{ 1 - \frac{1}{3}\alpha ; \frac{5}{3}\alpha, \alpha, -1 \right\} \quad \square$$

$$S = \{ A \in \mathcal{M}_2(\mathbb{R}) \mid {}^tA = A \} \subset \mathcal{M}_2(\mathbb{R})$$

$\forall \alpha, \beta \in K \ \forall x, y \in V$

$\alpha, \beta \in \mathbb{R}, \forall x, y \in V, \alpha x + \beta y \in V$

$\alpha A + \beta B = (\alpha A + \beta B)^t$

$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$; ${}^tA = \begin{pmatrix} a & c \\ b & d \end{pmatrix}$

$A = {}^tA \Rightarrow b = c = b$

$$\begin{pmatrix} \alpha a & \alpha b \\ \alpha c & \alpha d \end{pmatrix} + \begin{pmatrix} \beta a_1 & \beta b_2 \\ \beta b_2 & \beta d \end{pmatrix} = \begin{pmatrix} \alpha a + \beta a_1 & \alpha b + \beta b_2 \\ \alpha b + \beta b_2 & \alpha d + \beta a \end{pmatrix}$$

S

$\Downarrow$

$O_2 \in S$

$\Rightarrow$ subspatiu

$D = \{ D \in \mathcal{M}_2(\mathbb{R}) \mid \exists \alpha \in \mathbb{R} \text{ a.î. } D = \alpha I_2 \} \subset \mathcal{M}_2(\mathbb{R})$

Fie $\left. D = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in D \right\} \Rightarrow D = \alpha I_2 \Rightarrow$

$\begin{pmatrix} a & b \end{pmatrix}$

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} = \alpha \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} \alpha & 0 \\ 0 & \alpha \end{pmatrix} \Rightarrow \begin{cases} a = d = \alpha \\ c = b = 0 \end{cases}$$

Deci $D = \begin{pmatrix} \alpha & 0 \\ 0 & \alpha \end{pmatrix}$

$$\left.\begin{array}{l} \forall\, V_1, V_2 \in D \\ \forall\, K_1, K_2 \in \mathbb{R} \end{array}\right\} \overset{?}{\Rightarrow} K_1 V_1 + K_2 V_2 \in D$$

Fie $V_1 = \begin{pmatrix} \alpha_1 & 0 \\ 0 & \alpha_1 \end{pmatrix} \in D$ și $V_2 = \begin{pmatrix} \alpha_2 & 0 \\ 0 & \alpha_2 \end{pmatrix} \in D$

$$K_1 V_1 + K_2 V_2 = K_1 \begin{pmatrix} \alpha_1 & 0 \\ 0 & \alpha_1 \end{pmatrix} + K_2 \begin{pmatrix} \alpha_2 & 0 \\ 0 & \alpha_2 \end{pmatrix}$$

$$= \begin{pmatrix} K_1\alpha_1 + K_2\alpha_2 & 0 \\ 0 & K_1\alpha_1 + K_2\alpha_2 \end{pmatrix} = (K_1\alpha_1 + K_2\alpha_2)\begin{pmatrix} 1 & 0 \end{pmatrix}$$

$$\left.\begin{array}{l} \left(\underbrace{k_1\alpha_1 + k_2\alpha_2}_{\in \mathbb{R}}\right) i_2 \\ 0_2 \in \mathcal{M}_2(\mathbb{R}) \end{array}\right\} \Rightarrow \text{subspatiu}$$

$$\left(\begin{array}{ccc|c} \alpha & 1 & 1 & 1 \\ 1 & \alpha & 1 & 1 \\ 1 & 1 & \alpha & 1 \end{array}\right) \overset{\substack{L_2 = L_1 \\ L_1 = L_2}}{=} \left(\begin{array}{ccc|c} 1 & \alpha & 1 & 1 \\ \alpha & 1 & 1 & 1 \\ 1 & 1 & \alpha & 1 \end{array}\right) \overset{\substack{L_2 = L_2 - \alpha L_1 \\ L_3 = L_3 - L_1}}{=}$$

$$\left(\begin{array}{ccc|c} \boxed{1} & \alpha & 1 & 1 \\ 0 & 1-\alpha^2 & 1-\alpha & 1-\alpha \\ 0 & 1-\alpha & \alpha-1 & 0 \end{array}\right) \overset{\substack{L_2 = L_3 \\ L_3 = L_2}}{=}$$

$$\left(\begin{array}{ccc|c} 1 & \alpha & 1 & 1 \\ 0 & 1-\alpha & \alpha-1 & 0 \end{array}\right) \overset{L_2 = \frac{1}{1-\alpha} L_2}{=}$$

$$\left(\begin{array}{ccc|c} 0 & 1-\alpha^2 & 1-\alpha & 1-\alpha \end{array}\right)$$

$$\left(\begin{array}{ccc|c} 1 & \alpha & 1 & 1 \\ 0 & \boxed{1} & -1 & 0 \\ 0 & 1-\alpha^2 & 1-\alpha & 1-\alpha \end{array}\right)$$

$L_3 = L_3 - (1-\alpha^2)L_2$

$L_1 = L_1 - \alpha L_2$

$(1-\alpha)(1-\alpha+1) = (1-\alpha)(-\alpha+2)$

$$\left(\begin{array}{ccc|c} 1 & 0 & 1+\alpha & 1 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & (1-\alpha)-(1-\alpha^2) & 1-\alpha \end{array}\right)$$

$L_3 = \dfrac{L_3}{(1-\alpha)(-\alpha+2)}$

$$\left(\begin{array}{ccc|c} 1 & 0 & 1+\alpha & 1 \end{array}\right)$$

$L_2 = L_3 + L_2$

$$\left(\begin{array}{ccc|c} 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & -\alpha+2 \end{array}\right) \quad L_1 = L_1 - L_3 - \alpha L_3$$

$$\left(\begin{array}{ccc|c} 1 & 0 & 0 & 1-2+\alpha+\alpha^2-\alpha 2 \\ 0 & 1 & 0 & -\alpha+2 \\ 0 & 0 & 1 & -\alpha+2 \end{array}\right) \quad \forall \alpha \in \mathbb{R} \setminus \{-1;2\}$$

$\square$

$$\begin{cases} \alpha x + y + z = 1 \\ x + \alpha y + z = \alpha \\ x + y + \alpha z = \alpha^2 \end{cases}$$

→ se vor efectua aceiași pași de mai sus, doar că rezultatul final va fi diferit. Voi scrie doar rezultatele finale (de după bară)

$$\begin{array}{l} L_2 = L_1 \\ L_1 = L_2 \end{array} \Rightarrow \left(\begin{array}{c} \alpha \\ 1 \\ \alpha^2 \end{array}\right)$$

$$\begin{array}{l} L_2 = L_3 \\ L_3 = L_2 \end{array} \Rightarrow \left(\begin{array}{c} 1-\alpha^2 \\ \alpha \end{array}\right)$$

$L_2 = L_2 - \alpha L_1$

$$L_3 = L_3 - L_1 \Rightarrow \begin{pmatrix} 1-\alpha^2 \\ \alpha(\alpha-1) \end{pmatrix} \qquad \begin{pmatrix} \alpha(\alpha-1) \end{pmatrix}$$

$$L_2 = \frac{L_2}{1-\alpha} \Rightarrow \begin{pmatrix} 1-\alpha^2 \\ \frac{\alpha}{1-\alpha} \\ \alpha(\alpha-1) \end{pmatrix}$$

$$\begin{aligned} L_3 &= L_3 - (1-\alpha^2)L_2 \\ L_1 &= L_1 - \alpha L_2 \end{aligned} \Rightarrow \begin{pmatrix} 1-\alpha^2 - \frac{\alpha^2}{1-\alpha} = \frac{1}{(1-\alpha)^3} \\ \frac{\alpha}{1-\alpha} \\ \underbrace{\alpha(\alpha-1)}_{\alpha^2-\alpha} - \underbrace{(1+\alpha)\alpha}_{\alpha^2+\alpha} = 0 \end{pmatrix} = \begin{pmatrix} \frac{1}{(1-\alpha)^3} \\ \frac{\alpha}{1-\alpha} \\ 0 \end{pmatrix}$$

$$L_3 = \frac{L_3}{-(\alpha-2)(\alpha+1)} \Rightarrow \begin{pmatrix} \left(\frac{1}{1-\alpha}\right)^3 \\ \frac{\alpha}{1-\alpha} \\ 0 \end{pmatrix}$$

$L_1 = L_1 - L_3 - \alpha L_3 \quad \Rightarrow \quad \begin{pmatrix} \left(\frac{1}{1-\alpha}\right)^3 \\ \frac{\alpha}{1-\alpha} \\ 0 \end{pmatrix}$

$\forall \alpha \in \mathbb{R} \setminus \{-1; 2\}$

Resultat final : $\left(\begin{array}{ccc|c} 1 & 0 & 0 & \left(\frac{1}{1-\alpha}\right)^3 \\ 0 & 1 & 0 & \frac{\alpha}{1-\alpha} \\ 0 & 0 & 1 & 0 \end{array}\right)$ $\square$

$$A = \begin{pmatrix} 1 & 1 & 2 & 3 \\ 1 & 1 & 3 & 4 \\ 2 & 5 & 1 & -1 \\ -1 & -2 & 2 & 4 \end{pmatrix}$$

a) L_3

$\det = (-1)^{3+1} \cdot 2 \begin{vmatrix} 1 & 2 & 3 \\ 1 & 3 & 4 \end{vmatrix} + (-1)^{3+2} \cdot 5 \begin{vmatrix} 1 & 2 & 3 \\ 1 & 3 & 4 \end{vmatrix} +$

$$\begin{vmatrix} 1 & 3 & 4 \\ -2 & 2 & 4 \end{vmatrix} \qquad \begin{vmatrix} 1 & 3 & 4 \\ -1 & 2 & 4 \end{vmatrix}$$

$$(-1)^{3+3} \cdot 1 \begin{vmatrix} 1 & 1 & 3 \\ 1 & 1 & 4 \\ -1 & -2 & 4 \end{vmatrix} + (-1)^{3+4} \cdot (-1) \begin{vmatrix} 1 & 1 & 2 \\ 1 & 1 & 3 \\ -1 & -2 & 2 \end{vmatrix} =$$

$$2(12 + 6 - 16 + 18 - 8 - 8) - 5(12 + 6 - 8 + 9 - 8 - 8) +$$

$$(4 - 6 - 4 + 3 + 8 - 4) + (2 - 4 - 3 + 2 + 6 - 2) =$$

$$2 \cdot 4 - 5 \cdot 3 + 1 + 1 = -5$$

b) după ultimele 2 linii

$$(-1)^{3+4+1+2} \begin{vmatrix} 2 & 5 \\ -1 & -2 \end{vmatrix} \begin{vmatrix} 2 & 3 \\ 3 & 4 \end{vmatrix} + (-1)^{3+4+1+3} \begin{vmatrix} 2 & 1 \\ -1 & 2 \end{vmatrix} \begin{vmatrix} 1 & 3 \\ 1 & 4 \end{vmatrix} +$$

$$(-1)^{3+4+1+4} \begin{vmatrix} 2 & -1 \\ -1 & 4 \end{vmatrix} \begin{vmatrix} 1 & 2 \\ 1 & 3 \end{vmatrix} + (-1)^{3+4+2+3} \begin{vmatrix} 5 & 1 \\ -2 & 2 \end{vmatrix} \begin{vmatrix} 1 & 3 \\ 1 & 4 \end{vmatrix} +$$

$$(-1)^{3+4+2+4} \begin{vmatrix} 5 & \ldots \end{vmatrix}$$

$$(-1)\begin{vmatrix} 5 & -1 \\ -2 & 4 \end{vmatrix} + (-1)^{3+4+3+2}\begin{vmatrix} 1 & -1 \\ 2 & 4 \end{vmatrix}\begin{vmatrix} 1 & 1 \\ 1 & 1 \end{vmatrix}$$

$$= (-4+5)(8-9) - (4+1)(4-3) + (8+1)(3-2) +$$

$$(10+2)(4-3) - (20+2)(3-2) + 0 = +1-5+9+12-22$$

$$= -5 \quad \square$$

SLD

$v_1 = (1,3,5)$ $\quad v_2 = (2,-1,4)$ $\quad v_3 = (8,3,22)$

$$\begin{pmatrix} 1 & 2 & 8 \\ 3 & -1 & 3 \\ 5 & 4 & 22 \end{pmatrix} \quad \begin{matrix} L_2 = L_2 - 3L_1 \\ L_3 = L_3 - 5L_1 \end{matrix}$$

L_2 și L_3 sunt proporționale

$$\begin{pmatrix} 1 & 2 & 0 \\ 0 & -7 & -21 \\ 0 & -6 & -18 \end{pmatrix} \Rightarrow \text{SLD}$$

$\square$

SLi

$v_1 = (1,3,2)$, $v_2 = (2,-1,1)$, $v_3 = (0,4,a)$

$$\begin{pmatrix} 1 & 2 & 0 \\ 3 & -1 & 4 \\ 2 & 1 & a \end{pmatrix} \begin{matrix} L_3 = L_3 - 2L_1 \\ L_2 = L_2 - 3L_1 \\ = \end{matrix}$$

$$\begin{pmatrix} \boxed{1} & 2 & 0 \\ 0 & \boxed{-7} & 4 \\ 0 & -3 & a \end{pmatrix} \begin{matrix} L_1 = L_1 + \frac{2}{7}L_2 \\ L_3 = L_1 - \frac{3}{7}L_2 \\ = \end{matrix} \begin{pmatrix} 1 & 0 & \frac{8}{7} \\ 0 & -7 & 4 \\ 0 & 0 & \boxed{-\frac{12}{7} + a} \end{pmatrix}$$

$L_2 = L_2 -$

$$\frac{28}{7} = \frac{-12+7a}{7} \Rightarrow 28 = -12 + 7a, \quad a = \frac{40}{7} \Rightarrow a = \mathbb{R} \setminus \left\{\frac{20}{7}, \frac{40}{7}\right\}$$

$$\frac{8}{7} = -\frac{12}{7} + a \Rightarrow a = \frac{20}{7}$$

$m, n \in \mathbb{N}^*$. Dim K-sp vect $M_{m,n}(K)$ este mn.
(K este subcorpul lui $\mathbb{C}$)

Fie $A \in M$

$$\begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ \vdots & & & \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix}$$

mărimea este $m \cdot n$

$$\underbrace{V_1 \quad V_2 \quad \dots \quad V_n}_{n \text{ vectori de mărime } m}$$

$\square$

K subcorp al lui $\mathbb{C}$, $n \in \mathbb{N}^*$

$\dim K[X]_n = n+1$

rezultă din condiția impusă, iar 1 este la elem neutru

$\dim_{\mathbb{C}} \mathbb{C} = 1$ $\dim_{\mathbb{R}} \mathbb{C} = 2$, iar $\mathbb{Q}$ sp vect $\mathbb{C}$ nu este finit generat $\square$

$\mathbb{C} = \mathbb{C} \Rightarrow \dim_{\mathbb{C}} \mathbb{C} = 1$

Fie x, y elem din $\mathbb{R}$

Un element din $\mathbb{C} = x + iy$

$\Rightarrow \mathbb{C} = \mathbb{R} \times \mathbb{R} \Rightarrow \dim_{\mathbb{R}} \mathbb{C} = 2$

partea reală partea imaginară

Pt $\mathbb{Q}$ nu se poate face asta, deoarece nu știm exact câte elemente are în plus $\mathbb{R}$ față de $\mathbb{Q}$ $\square$

$v_1 = (1,1,1) \qquad v_2 = (2,3,-1) \qquad v_3 = (4,5,1)$

$U = \langle v_1, v_2, v_3 \rangle$ dim + baza

$$\begin{vmatrix} 1 & 2 & 4 \\ 1 & 3 & 5 \\ 1 & -1 & 1 \end{vmatrix} = 3 - 4 + 10 - 12 + 5 - 2 = 0$$

$$\begin{vmatrix} 1 & 2 \\ 1 & 3 \end{vmatrix} = 1 \neq 0 \Rightarrow \text{rang } U = 2 \Rightarrow \dim U = 2$$

Fie $z = \alpha$

$$\begin{aligned} x + 2y &= -4\alpha \\ x + 3y &= -5\alpha \\ x - y &= -\alpha \end{aligned} \Rightarrow \begin{aligned} y &= -\alpha \\ x &= -2\alpha \end{aligned}$$

$\Rightarrow \alpha(-2, -1, 1)$

$\Rightarrow U = \langle \{ (-2, -1, 1) \} \rangle$

bază în U $\square$

$U = \{ (x+3y+5z,\ x-y+z,\ -x+2y,\ x+y+3z) \}$

U ssp $\mathbb{R}^4$ dim U + bază

$$\begin{pmatrix} 1 & 3 & 5 & 0 \\ 1 & -1 & 1 & 0 \\ -1 & 2 & 0 & 0 \\ 1 & 1 & 3 & 0 \end{pmatrix} \qquad \begin{vmatrix} 1 & 3 & 5 \\ 1 & -1 & 1 \\ -1 & 2 & 0 \end{vmatrix} = \qquad 10-3-5-2=0$$

$\Rightarrow$ rang $U = 2$

deoarece $t = 0$

$z = \alpha$ nec nec

$\Downarrow$

$\dim U = 2$

$$\left.\begin{aligned} x+3y &= -5\alpha \\ x-y &= -\alpha \\ -x+2y &= 0 \\ x+y &= -3\alpha \end{aligned}\right\} \Rightarrow \left.\begin{aligned} y &= -\alpha \\ x &= -2\alpha \end{aligned}\right\} \Rightarrow \alpha(-2; -1; 1; 0)$$

$\Downarrow$

$U = \mathcal{L}\{(-2;1;1;0)\}$

bază $\square$

$f(x,y,z) = (2x - y + 2z, -x + 2y - z, x + y + z)$

$\operatorname{Ker} f + \operatorname{Im} f$ inj, surj, bij

spectrul endomorfismului și subsp propriu

endomorfism diagonalizabil

determinați A^n, $n \in \mathbb{N}^*$

$$\det f = \begin{vmatrix} 2 & -1 & 2 \\ -1 & 2 & -1 \\ 1 & 1 & 1 \end{vmatrix} = 4 - 2 + 1 - 4 + 2 - 1 = 0$$

$\Downarrow$

sol nulă nu este unică

$z = \alpha$

$2x - y = -2\alpha$

$-x + 2y = \alpha \Rightarrow x = -\alpha$

$x+y=-\alpha \qquad y=0$

$\operatorname{Ker} f = \{ 0_{\mathbb{R}^3}, \alpha(-1,0,1) \} \Rightarrow \dim_{\mathbb{R}} \operatorname{Ker} f = 1$

$\operatorname{Im} f = 3-1=2$ (conform Th. Rang Defect)

f nu este nici inj și nici surj

$$\det(A-\lambda I_3) = 0 \Leftrightarrow \begin{vmatrix} 2-\lambda & -1 & 2 \\ -1 & 2-\lambda & -1 \\ 1 & 1 & 1-\lambda \end{vmatrix}$$

$$(2-\lambda)^2(1-\lambda) - 2 + 1 - 4 + 2\lambda + 2 - \lambda - 1 + \lambda = 0$$

$$(2-\lambda)^2(1-\lambda) - 4 + 2\lambda = 0$$

$$(2-\lambda)^2(1-\lambda) - 2(2-\lambda) = 0$$

$$(2-\lambda)\big((2-\lambda)(1-\lambda)-2\big)=0$$

$$(2-\lambda)(2-3\lambda+\lambda^2-2)=0$$

$$\lambda(2-\lambda)(-3+\lambda)=0$$

$\operatorname{spec}(f)=\{0,2,3\}$ valori proprii

$\operatorname{ma}(\lambda_i)=1,\ (\forall)\quad i=\{1,2,3\}$

ssp proprii : $V_{\lambda_1}, V_{\lambda_2}, V_{\lambda_3}$

$$V_{\lambda_1}:\quad \begin{pmatrix} 2 & -1 & 2 \\ -1 & 2 & -1 \\ 1 & 1 & 1 \end{pmatrix}\begin{pmatrix} x \\ y \\ z \end{pmatrix}=\begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$\text{rang}(A + i_3) = 2$

$\alpha(-1, 0, 1) = \langle v_1 \rangle$

$B_1 = \{v_1\} \subset V_{\lambda_1} \Rightarrow \dim V_{\lambda_1} = 1 \Rightarrow mg(\lambda_1) = 1$

baza

V_{λ_2} :

$$\begin{pmatrix} 0 & -1 & 2 \\ -1 & 0 & -1 \\ 1 & 1 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$\det = 0 - 2 + 1 + 1 = 0 \Rightarrow \text{rang}(A + i_3) = 2$

$z = \alpha$ nec sec.

$$\begin{cases} -x = \alpha \\ x + y = +\alpha \end{cases} \Rightarrow \alpha(-1, 2, 1) = \langle v_2 \rangle$$

$\dim V_{\lambda_2} = 1$

$B_{\lambda_2} = \{v_2\} \subset V_{\lambda_2} \Rightarrow \dim V_{\lambda_2} = 1 \Rightarrow mg(\lambda_2) = 1$

bazza

$$\begin{vmatrix} -1 & -1 & 2 \\ -1 & -1 & -1 \\ 1 & 1 & -2 \end{vmatrix} = -2 - 2 + 1 + 2 - 1 + 2 = 0$$

$\operatorname{rang}(A + i3) = 2$

$x = \alpha$ nec sec

$$\begin{cases} -x + 2z = \alpha \\ -y - z = \alpha \\ y - z = -\alpha \end{cases} \qquad \begin{cases} z = 0 \\ y = -\alpha \end{cases} \Rightarrow \alpha(1, -1, 0) = \langle v_3 \rangle$$

$B_{\lambda_3} = \{v_3\} \subset V_{\lambda_3} \Rightarrow \dim V_{\lambda_3} = 1 \Rightarrow mg(\lambda_3) = 1$

$U_3 = \{v_3\} \subset V_{\lambda_3} \Rightarrow \dim V_{\lambda_3} = \dots = mg(\lambda_3)$

Rezultă

$$ma(\lambda_1) + ma(\lambda_2) + ma(\lambda_3) = 3 = \underbrace{\dim_{\mathbb{R}} \mathbb{R}^3}_{}$$

$$ma(\lambda_i) = mg(\lambda_i) = 1, \ \forall i = \overline{1,3}$$

$\Rightarrow f$ endom. diag

$$A = \begin{pmatrix} 2 & -1 & 2 \\ -1 & 2 & -1 \\ 1 & 1 & 1 \end{pmatrix} \qquad [f]_{BO_3} = A$$

f endom diag $\Rightarrow (\exists)\ \tilde{B} \subset \mathbb{R}^3$ a.î. $[f]_{\tilde{B}} \to$ m diag

$$D = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix} \qquad BO_3 \underset{\mathcal{C}}{\sim} \tilde{B} = \{v_1, v_2, v_3\}$$

$\downarrow \qquad \downarrow$

$$\tilde{B} = B_1 \cup B_2 \cup B_3$$
$$= \{v_1, v_2, v_3\}$$

$$A^n = C \cdot D \cdot C^{-1}$$

$$D^n = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 2^n & 0 \\ 0 & 0 & 3^n \end{pmatrix}$$

A

$$D = C A C \Big| \begin{matrix} C \text{ hat} \\ C^{-1} \text{ hat} \end{matrix}$$

$$A = C \cdot D \cdot C^{-1}$$